AD 2. AERODROMES

VEBS AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VEBS - BIJU PATNAIK INTERNATIONAL AIRPORT, BHUBANESWAR

VEBS AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	201448N 0854907E MAG. BRG. 343.00 DEG / 660 M from physical Extremity of RWY 32.	
2	Direction and distance of aerodrome reference point from the centre of the city or town which the aerodrome serves	BRG 224 DEG/3 KM from Bhubaneswar Railway station.	
3	Aerodrome elevation and reference temperature	141 FT / 37.7 DEG C	
4	Magnetic variation, date of information and annual change	0.67 DEG W (2010) /0.00	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	The Airport Director Airports Authority of India, Biju Patnaik Airport, Bhubaneswar - 751020	
		Telephone:	(91) 6742596300 (91) 9437496302
		Fax:	(91) 6742596302
		AFS:	VEBSYHYX
		Email:	apdbbsr@AAI.AERO
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	Elevation in EGM 08	

VEBS AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1230 UTC (0930-1800 IST) SAT, SUN+ HOL : Nil	
2	Custom and immigration	H24	
3	Health and sanitation	NIL	
4	AIS briefing office	H24	
5	ATS reporting office (ARO)	H24	
6	MET Briefing office	H24	
7	Air Traffic Service	Consult NOTAM for Current ATS HR.	
8	Fuelling	H24	
9	Handling	As ATS	
10	Security	H24	
11	De-icing	NIL	
12	Remarks	Outside of NOTAM/ATS hours, services are available on request with 24 HR prior notice.	
		ATS approved hourly runway traffic handling capacity	Maximum number of arrival and departure: 12 Maximum number of arrival only: 06 Maximum number of departure only -10

VEBS AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	NIL
2	Fuel and Oil types	JET A1 AVGAS, 100LL Oil MJO II
3	Fuelling facilities and capacity	Reliance: 5 Refuellers:16000 L (2), 11000 L (2), Preset: 700 -750 LPM, 6000L Preset: 400 LPM HPCL: 2 Refuellers: 12000L (1), 15000L (1) Preset: 1200 LPM
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	NIL

VEBS AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city.
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Buses, Taxi and Car Hire from AD.
4	Medical Facilities	First aid at AD. Hospital in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: In the city. SBI ATM at AD. Post office: within AD ADMN HR.
6	Tourist office	At AD
7	Remarks	NIL

VEBS AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-7
2	Rescue equipment	3 Ambulances
3	Capability for removal of disabled aircraft	1. The critical Aircraft for the Airport is A321. 2. Primary responsibility for removal of disabled aircraft removal rests with the concerned airline. 3. The airlines may contact ‘Aerodrome Coordinator for Disabled Aircraft Removal Operations’ for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: AD Coordinator- Airport Director Contact No.: 9437496302, 0674-2596300 (O)/ 2596301 (R)/ Fax-02652464249 Email: apdbbsr@aai.aero
4	Remarks	NIL

VEBS AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VEBS AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking / Docking Charts for Details
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking / Docking Charts for Details
3	Location and elevation of altimeter checkpoints	Location: At Apron 201452.2N 0854905.7E Elevation: 144 FT
4	Location of VOR checkpoints	On TWY B: 201503.06N 0854850.73E ELEV: 135 FT
5	Position of INS checkpoints	
6	Remarks	**1. Coordinates of RWY Holding Position** **TWY COORDINATES ELEVATION(FT)** TWY B 201502.82N 0854850.39E 134 TWY C 201510.04N 0854844.79E 136 TWY D 201458.83N 0854853.51E 133 TWY J 201540.64N 0854821.02E 106 (with RWY 14) TWY J 201429.13N 0854916.23E 128 (with RWY 32) **2. Coordinates of RWY & Taxi Intersection** **TWY COORDINATES ELEVATION(FT)** TWY B 201501.02N 0854847.76E 135 TWY C 201508.24N 0854842.16E 134 TWY D 201457.05N 0854850.83E 136 TWY J 201538.86N 0854818.46E 105 (with RWY 14) TWY J 201428.00N 0854914.60E 130 (with RWY 32) 3.Markings: Apron edge and aircraft stand marking. 4. Lighting: Apron floodlights. 5. Ramp Equipment Area: 153.5MX10M provided on East Side of the apron. 6. Stands 1 to 6 are Power in Power out. 7. TWY C: Length - 229M & Shoulder - 7.5M. 8 TWY D: Length 166M capable up to Type B Aircraft. 9. Gravity Model: EGM08

VEBS AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiing guidance provided on RTF Taxi -In and Out markings available.

2	Runway and taxiway markings and lights	RWY Markings:RWY- Designation,Centerline, THR, TDZ,Edge, RWY End, Turn pad. RWY Lights: RWY Edge Lights, RWY End Lights, Threshold Lights,Turnpad lights. TWY Marking: Centreline, Edge, TWY Holding Position. TWY Lights: TWY Edge lights, (TWY A, TWY B, TWY C, TWY D, TWY J, TWY K TWY L and TWY R). IHP lights, Stop Bar lights, Guard lights, No Entry lights of RET.
3	Stop bars (if any)	RWY Holding at TWY J (Both RWY 14 and RWY 32), TWY B, TWY C
4	Remarks	Guard Lights available At Runway holding position TWY J (Both RWY 14 and RWY 32), TWY B, TWY C Apron Edge Marking, Aircraft Stand Marking

VEBS AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
32/TKOF 14/APCH	TREE	201600.0N 0854757.6E	160 FT	NO	TREE
32/TKOF 14/APCH	TREE	201601.6N 0854759.2E	158 FT	NO	TREE (PIPAL)
32/TKOF 14/APCH	BUILDING	201601.4N 0854809.5E	149 FT	NO	BUILDING
32/TKOF 14/APCH	TREE	201558.1N 0854810.0E	144 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201555.8N 0854809.9E	140 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201600.0N 0854759.5E	160 FT	NO	TREE
32/TKOF 14/APCH	TREE	201600.9N 0854800.6E	163 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	POLE	201603.3N 0854759.4E	169 FT	NO	LIGHT POLE
32/TKOF 14/APCH	TREE	201612.1N 0854752.0E	191 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	OTHER	201610.2N 0854751.3E	186 FT	NO	MAST ON BUILDING (BUILDING ELEV. 52.2 M)
32/TKOF 14/APCH	OTHER	201616.4N 0854748.4E	206 FT	NO	MAST
32/TKOF 14/APCH	TREE	201616.8N 0854745.0E	201 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	OTHER	201617.8N 0854745.0E	207 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201618.3N 0854744.6E	211 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201618.8N 0854731.1E	226 FT	NO	MAST ON BUILDING (BUILDING TOP 53.5M)
32/TKOF 14/APCH	OTHER	201623.3N 0854736.7E	242 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201630.7N 0854735.6E	244 FT	NO	MAST

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
32/TKOF 14/APCH	OTHER	201630.9N 0854735.7E	233 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201641.0N 0854746.4E	287 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201653.3N 0854731.5E	294 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201640.8N 0854740.0E	286 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201645.6N 0854740.5E	261 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201640.1N 0854744.9E	269 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201629.4N 0854739.9E	228 FT	NO	MAST
32/TKOF 14/APCH	OTHER	201635.7N 0854747.4E	253 FT	NO	MAST
32/TKOF 14/APCH	TREE	201617.3N 0854751.6E	197 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201614.6N 0854751.6E	193 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201611.2N 0854755.3E	188 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201614.4N 0854754.5E	188 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201618.3N 0854756.5E	195 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201607.6N 0854801.5E	178 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201604.0N 0854801.5E	157 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	BUILDING	201606.5N 0854808.1E	160 FT	NO	BUILDING
32/TKOF 14/APCH	TREE	201548.1N 0854804.4E	153 FT	NO	TREE
32/TKOF 14/APCH	TREE	201554.7N 0854807.9E	148 FT	NO	GROUP OF TREES
32/TKOF 14/APCH	TREE	201550.7N 0854801.2E	157 FT	NO	TREE
32/TKOF 14/APCH	OTHER	201608.0N 0854806.5E	164 FT	NO	SINTEX TANK ON BUILDING (BUILDING TOP = 48.6 M)
32/APCH 14/TKOF	TREE	201424.0N 0854923.3E	158 FT	NO	GROUP OF TREES
32/APCH 14/TKOF	POLE	201423.3N 0854922.5E	139 FT	NO	ELECTRIC POLE
32/APCH 14/TKOF	POLE	201422.8N 0854921.9E	138 FT	NO	ELECTRIC POLE
32/APCH 14/TKOF	TREE	201421.7N 0854920.7E	140 FT	NO	GROUP OF TREES
32/APCH 14/TKOF	OTHER	201420.0N 0854917.2E	140 FT	NO	ROD ON LOCALIZER HUT(HUT TOP 37.8M)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
32/APCH 14/TKOF	ANTENNA	201420.1N 0854917.1E	140 FT	NO	ANTENNA ON LOCALIZER HUT
32/APCH 14/TKOF	POLE	201420.9N 0854923.5E	142 FT	NO	ELECTRIC POLE
32/APCH 14/TKOF	OTHER	201421.5N 0854924.5E	149 FT	NO	SINTEX ON BUILDING (BUILDING TOP 43.8M)
32/APCH 14/TKOF	TREE	201420.5N 0854926.6E	161 FT	NO	GROUP OF TREES
32/APCH 14/TKOF	BUILDING	201420.7N 0854926.0E	147 FT	NO	BUILDING
32/APCH 14/TKOF	TREE	201420.9N 0854925.0E	164 FT	NO	COCONUT TREE
32/APCH 14/TKOF	OTHER	201419.3N 0854923.2E	144 FT	NO	SINTEX ON BUILDING (BUILDING TOP 41.5M)
32/APCH 14/TKOF	TREE	201416.9N 0854924.3E	155 FT	NO	GROUP OF TREES
32/APCH 14/TKOF	TREE	201415.8N 0854919.3E	155 FT	NO	COCONUT TREE
32/APCH 14/TKOF	OTHER	201419.4N 0854916.2E	141 FT	NO	LIGHTNING ARRESTER
32/APCH 14/TKOF	OTHER	201425.5N 0854915.3E	126 FT	NO	APPROACH LIGHT
In circling area and at AD	OTHER	201535.9N 0854825.9E	122 FT	NO	GP MONITOR ANTENNA
In circling area and at AD	OTHER	201534.2N 0854827.3E	161 FT	NO	GP MAIN ANTENNA
In circling area and at AD	OTHER	201533.9N 0854827.4E	131 FT	NO	GP DME ANTENNA
In circling area and at AD	ANTENNA	201533.8N 0854827.5E	130 FT	NO	ANTENNA ON GP HUT
In circling area and at AD	OTHER	201532.8N 0854827.9E	116 FT	NO	RVR CAMERA
In circling area and at AD	OTHER	201532.6N 0854827.4E	117 FT	NO	SECURITY HUT
In circling area and at AD	ANTENNA	201532.0N 0854828.5E	143 FT	NO	MET ANTENNA
In circling area and at AD	OTHER	201531.9N 0854828.4E	116 FT	NO	RVR CAMERA
In circling area and at AD	OTHER	201527.7N 0854823.2E	134 FT	NO	WIND DIRECTION INDICATOR
In circling area and at AD	OTHER	201528.6N 0854821.2E	117 FT	NO	NEW GP HUT
In circling area and at AD	OTHER	201528.8N 0854821.1E	159 FT	NO	NEW GP MAIN ANTENNA
In circling area and at AD	OTHER	201530.4N 0854819.7E	122 FT	NO	NEW GP MONITOR ANTENNA
In circling area and at AD	OTHER	201528.6N 0854821.0E	122 FT	NO	NEW GP DME

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	201534.0N 0854827.4E	129 FT	NO	GP DME ANTENNA OLD
In circling area and at AD	OTHER	201433.7N 0854903.0E	156 FT	NO	WIND DIRECTION INDICATOR
In circling area and at AD	OTHER	201436.3N 0854910.9E	136 FT	NO	SECURITY HUT
In circling area and at AD	TREE	201527.5N 0854819.1E	149 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	201549.2N 0854800.0E	157 FT	NO	TREE
In circling area and at AD	BUILDING	201544.1N 0854704.1E	346 FT	NO	KHANDAGIRI TEMPLE
In circling area and at AD	OTHER	201655.3N 0854631.0E	350 FT	NO	OVER HEAD WATER TANK
In circling area and at AD	OTHER	201647.4N 0854754.4E	294 FT	NO	MAST
In circling area and at AD	OTHER	201740.1N 0854721.0E	329 FT	NO	PYLON MAST
In circling area and at AD	OTHER	201744.5N 0854729.6E	334 FT	NO	PYLON MAST
In circling area and at AD	OTHER	201507.3N 0854847.3E	146 FT	NO	SECURITY HUT
In circling area and at AD	OTHER	201503.2N 0854841.6E	153 FT	NO	WIND DIRECTION INDICATOR
In circling area and at AD	BUILDING	201610.7N 0854939.3E	310 FT	NO	BUILDING
In circling area and at AD	BUILDING	201546.2N 0854555.6E	315 FT	NO	MULTI STOREY BUILDING
In circling area and at AD	BUILDING	201352.3N 0854648.1E	322 FT	NO	MULTI STOREY BUILDING

VEBS AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Bhubaneswar
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Bhubaneswar Issued 09 Hourly Validity TAF for National at every Three Hour interval and 30 Hourly validity TAF for International at 1100 UTC on Sunday, Monday, Wednesday and Friday.
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend 30 MIN
5	Information on how briefing and/or consultation is provided	Yes, Briefing through Documentation and OLBS
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular Form (English)

7	Charts and other information displayed or available for briefing or consultation	1. WAFC Products: Data for upper wind & temperature for levels: 850 h.Pa, 700 h.Pa, 500 h.Pa, 300 h.Pa, 250 h.Pa, 200 h.Pa, 150 h.Pa, SIG Wx charts for FL 100 - 250 & for FL 250 - 630. 2. National Charts for consultation only. Surface charts, synoptic upper air charts for levels: 100, 150, 200, 250, 300, 400, 500, 600, 700, 850 hPa, prognostic upper air charts at constant pressure, prognostic significant weather chart, DWR products / Satellite imagery.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	RS / RW Ascent, MFI, HPCS, (Email, Fax, IMD website: http://imd.gov.in)
9	The air traffic services unit(s) provided with meteorological information	Bhubaneswar ATS
10	Additional information, e.g. concerning any limitation of service.	(i) Visibility and RVR assessed manually (ii) No Ceilometer installed (iii) DWR Paradeep Products are used for forecasting (iv) Transmissometer installed and commissioned for touch down zone of RWY 14.

VEBS AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
14	143.85 DEG	2744 x 45 M	828/F/C/W/T Asphalt	THR: 201539.18N 0854818.22E RWY END: 201427.12N 0854914.00E
32	323.85 DEG	2744 x 45 M	828/F/C/W/T Asphalt	THR: 201427.12N 0854914.00E RWY END: 201539.18N 0854818.22E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 105.0FT TDZ: 128.0FT	0.53%	NIL	NIL	2864 x 280 M
THR: 125.0FT TDZ: 141.0FT	0.53%	NIL	NIL	2864 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
165M x 90M		NIL	i. Elevations in EGM08
210M x 90M		NIL	i. Elevations in EGM08

VEBS AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
14	2744	2744	2744	2744	1 : 50
32	2744	2744	2744	2744	1 : 50

VEBS AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
14	CAT I 789 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (16.39M)	NIL
32	210 M BALS LIH	Green	PAPI LEFT/3.00 DEG MEHT (15.88M)	NIL

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
	2744 M 60 M White	Red	NIL	NIL
	2744 M 60 M White	Red	NIL	NIL

VEBS AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At Tower Building, FLG W&G EV2SEC As ATS HR
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Available and lighted
		Anemometer	On tower building not lighted
3	Taxiway edge and taxiway centre line lights;	Edge	TWY A, B, C, D, J, R, K, L, F and G.
		Centre Line	NIL
4	Secondary power supply including switch-over time;	Secondary Power supply to all lighting. Switch-over time: within 15 Sec.Lights associated with Runway and taxiway switch over time 1 sec	
5	Remarks	Apron Flood Lights AVBL	

VEBS AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not established
2	TLOF and/or FATO area elevation:	Not established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not established
4	True bearings of FATO;	Not established
5	Declared distances available	Not established
6	Approach and FATO lighting;	Not established
7	Remarks	Not established

VEBS AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 201439N 0854847E within a 30NM radius.
2	Vertical limits	FL 50
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Bhubaneswar Tower / Bhubaneswar Approach, English
5	Transition altitude	4000 FT
6	Hours of applicability	H24
7	Remarks	Caution for local (powered and glider) training flights in the AD circuit.

VEBS AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
APP	Bhubaneswar Approach	125.525 MHZ	
TWR	Bhubaneswar Tower	118.150 MHZ	
TWR	Bhubaneswar Tower	125.150 MHZ	
ATIS	Bhubaneswar Information	126.800 MHZ	
ALRS	Emergency Frequency	121.500 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	Refer NOTAM	MAIN FREQ.
	HO	NIL
	HO	BACKUP FREQ. FOR TWR AND APP.
	H24	NIL
	H24	NIL

VEBS AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 14 CAT I	IBHR	109.100 MHz	H24
GP 14		331.400 MHz	H24
DME ILS 14	IBHR	CH28X	H24
DVOR/DME	BBS	113.500 MHz CH82X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
201421.0N 0854918.8E			Identification: IBHR
201528.8N 0854821.1E			Glide Angle 3 Deg
201528.6N 0854821.0E	122 FT		Elevation in EGM 08
201439.2N 0854846.8E	158 FT		1. Vertical datum: EGM08

VEBS AD 2.20 LOCAL AERODROME REGULATIONS

• No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

VEBS AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

VEBS AD 2.22 FLIGHT PROCEDURES

1. ADS-B based ATS Surveillance services are provided to suitably equipped aircraft BTN 0230-0830 UTC on opportunity basis within the coverage area of Bhubaneswar ADS-B sensors on VHF 125.525 MHz in jurisdiction of Bhubaneswar APCH Control.
2. ADS-B equipped aircraft flying within the jurisdiction of Bhubaneswar APCH Control shall comply with relevant provisions.
3. In ADDN to ABV, Service may be provided at other times also for ATCO training.

VEBS AD 2.23 ADDITIONAL INFORMATION

1. Additional Wind sock commissioned at a distance of 200M and BRG 300 DEG from ARP. Three (03) number of windsocks are available and all are lighted.
2. LOC Tx-2 clearance restriction upto 32 DEG lifted. GP restricted for use upto 150 ft.
3. Aircraft stands details:

Aircraft Stand	Co-ordinate	Critical Aircraft	PCN
1	201506.80N 0854857.70E	B767-400	74/R/C/W/T
2	201509.20N 0854855.80E	B767-400	74/R/C/W/T
3	201511.70N 0854853.90E	A321	74/R/C/W/T
4	201512.80N 0854853.00E	A321	74/R/C/W/T
5	201513.80N 0854852.20E	A321	74/R/C/W/T
6	201514.90N 0854851.30E	A321	74/R/C/W/T

4. ADS-B Ground Equipment Commissioned and Operational.

5. LOW VISIBILITY PROCEDURES

5.1. Definitions:

5.1.1 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: as per Para5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.

5.1.2 Low Visibility Take-off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400m.

5.1.3 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

5.1.4 Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

5.1.5 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a. Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.

b. landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c. Landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

5.1.6 Visibility - Visibility for aeronautical purposes is the greater of:

a. the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b. The greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background.

Note 1. — The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2. — The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

5.2. General:

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

5.3. Minimum Requirements:

The following aeronautical ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures.

a. Runway edge lights,

b. Runway end lights,

c. Real time TDZ RVR.

d. Standby Power supply to maintain switch over time of 1 Second for Runway Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.

Note: - At Bhubaneswar Airport RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set) before the implementation of LVP

5.3.1 Un-serviceability of Aeronautical Ground Lights/Equipment before Implementation of LVP

Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 4 above is un-serviceable or is not maintained as per the required standard.

5.3.2 Un-serviceability of Aeronautical Ground Lights/ Equipment after Implementation of LVP.

When any of the light/equipment mentioned in para 4 above becomes un-serviceable or fails to meet the required standard during periods of LVP, TWR shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

5.4. Safeguarding Procedures:

Safeguarding Procedures (SP) are instructions for relevant airport, departments and airside operators to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all Safeguarding procedures are complete. Duty Officer Tower/Tower Supervisor will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

5.4.1 Safeguarding Procedures shall be initiated when:

a. The Visibility/RVR is less than 1200m and visibility/RVR is forecast to deteriorate to 800m or less; and/or

b. The cloud ceiling is less than 400ft and forecast to fall to 200ft or less.

5.4.2 Safeguarding procedures include:

a. Positioning of 1 CFT each at the two predetermined positions near DVOR approach road and on fire station link road.

b. Stopping of all maintenance works on the maneuvering area as well as removal of all men and mobile equipment from the said area.

c. Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge and RWY End lights supported by DG SET.

(Note: RWY Edge and RWY End lights may continue to operate on main power supply during safe Guarding Procedures. Whenever, LVP is to be implemented as per para 6 below, the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented.

In case of DG set, Main power supply shall act as standby power. At airports where UPS is capable of maintaining the required AGL system with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Standby Power supply.)

d. The appropriate aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two-hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.

5.5. Low Visibility Procedures:

5.5.1 Implementation of Low Visibility Procedures:

Duty MET Officer shall inform Tower Supervisor whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below. Tower Supervisor shall coordinate with all the agencies to confirm whether the safeguarding procedures have been completed or not. When Visibility/RVR falls below 800m/550M and or Cloud Ceiling is 200 ft or below and safeguarding procedures are complete, Tower supervisor will implement Low Visibility Procedures. Duty Officer Tower/Tower Supervisor shall inform all users of the imposition of low visibility procedures

5.5.2 Action by various units during LVP:

a. Duty MET Officer shall keep Duty Officer Tower/Tower Supervisor informed of any change in Visibility/RVR.

b. TOWER shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. "Follow Me" shall inform ATC of any deviation from the route cleared by ATC;

c. Tower shall not permit any ground run on the manoeuvring area except idle power run on the stands;

d. Tower shall ensure that "Follow Me" services are provided to pilots on request;

e. The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all operating on the manoeuvring area shall be maintained by TOWER /OPS through GFS.

f. The following may be included in DATIS. **"Low Visibility Procedures in Force."**

g. TWR shall permit departures only from the beginning of the Runway in use.

h. Duty Officer Tower/Tower Supervisor shall stop all flight operations whenever RVR falls below 400M.

i. In-Charge Electrical shall continuously monitor the main and Standby Power supply to ensure changeover time of maximum one second for RWY Edge and RWY End lights during low visibility operations and report any un-serviceability to Tower immediately.

5.6 Termination of Low Visibility Procedures:

a. When Visibility/RVR improves to 800M or more and cloud ceiling is 200 feet or Higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.

b. The Tower Supervisor will intimate GFS/OPS/ARFF/ In-Charge Electrical- Engineering regarding the termination of LVP operations.

c. On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. **"LOW VISIBILITY PROCEDURES CANCELLED".**

d. If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, Tower Supervisor/Duty Officer Tower may cancel SP.

5.7. Actions by other Agencies (Airlines, Refuelling Companies, Catering Agencies, etc.):

a. Every Year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b. All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations.

c. A refresher program for ATCO's and personnel responsible for airside operations shall be conducted every year.

d. All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

e. All the vehicles must have their obstruction lights "ON" during Low Visibility Procedures operations.

f. Everybody must follow all instructions/sign boards provided for vehicular movement area/service roads.

6. Bird activity observed in and around airfield.Pilot to exercise caution while landing and takeoff.

7. Following intersection Take Off Run Available :

RWY	FROM TWY INT	TORA(M)
RWY 14	TWY B	1291
RWY 14	TWY C	1566
RWY 32	TWY F	2238
RWY 32	TWY D	1603

8.RWY turn pad capable of handling B777-300 ER type of ACFT, provided at both ends of RWY 14/32 for special OPS with prior permission.

9. ASMGCS commissioned to enhance visual surveillance.

10. Sectorwise slope of RWY 32

Longitudinal Slope of RWY (sector wise) from RWY 32 side

Section	Slope
1. From 0 M to 152.00 M	(+) 0.63%
2. From 152.00 M to 620.00 M	(+) 0.80%
3. From 620.00 M to 1082.00 M	(-) 0.40%
4. From 1082.00 M to 1470.00 M	(-) 0.08%
5. From 1470.00 M to 1820.00 M	(-) 0.54%
6. From 1820.00 M to 2744.00 M	(-) 0.80%

VEBS AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart(14 MAY 2026)
2. Aircraft Parking/Docking Chart(14 MAY 2026)
3. Aerodrome Obstacle Chart Type-A (Obstacle Limitations) RWY 14
4. Aerodrome Obstacle Chart Type-A (Obstacle Limitations) RWY 32
5. ILS Procedure RWY 14
6. VOR (Z) Procedure RWY 14
7. VOR (Z) Procedure RWY 32
8. ATC Surveillance Minimum Altitude Chart
9. RNP Y Procedure RWY 14
10. VEBS-RNP-Y-RWY-14-CODING
11. RNP Y Procedure RWY 32
12. VEBS-RNP-Y-RWY-32-CODING

VEBS AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3